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پروژه نهایی درس طراحی کامپایلر

استاد صادقی

فهرست مطالب

3..... طراحی گرامر زبان

4..... ساختار فایل‌های پروژه

5..... نمونه اجرای برنامه

طراحی گرامر زبان

program $\rightarrow$ statements

statements $\rightarrow$ statements statement $| \epsilon$

statement $\rightarrow$ declaration $|$ assignment $|$ block $|$ if_statement

Type $\rightarrow$ mana $|$ elixir

Declaration $\rightarrow$ type ID $\sim$ $|$ type ID $\Rightarrow$ expression $\sim$

Assignment $\rightarrow$ ID $\Rightarrow$ expression $\sim$

Block $\rightarrow$ begin_spell statements end_spell

if_statement $\rightarrow$ cast_if (condition) block $|$ cast_if (condition) block cast_else block

condition $\rightarrow$ expression == expression

Expression $\rightarrow$ expression + expression $|$ expression – expression $|$ expression * expression $|$
expression / expression $|$ (expression) $|$ ID $|$ INTEGER $|$ FLOAT

ساختار فایل‌های پروژه

SpellCraft/

|

└─ lexer.l

└─ parser.y

└─ symbol_table.h

└─ symbol_table.c

└─ main.c

|

└─ parser.tab.c (Generated by Bison)

└─ parser.tab.h (Generated by Bison)

└─ lex.yy.c (Generated by Flex)

|

└─ spellcraft.exe (Executable)

نمونه اجرای برنامه

```
Olia@Olia MINGW64 /d/olia/desk/danesh ga/term 6/compiler/compiler_project_3rd_try
$ bison -d parser.y

Olia@Olia MINGW64 /d/olia/desk/danesh ga/term 6/compiler/compiler_project_3rd_try
$ flex lexer.l

Olia@Olia MINGW64 /d/olia/desk/danesh ga/term 6/compiler/compiler_project_3rd_try
$ gcc parser.tab.c lex.yy.c symbol_table.c main.c -o spellcraft.exe

Olia@Olia MINGW64 /d/olia/desk/danesh ga/term 6/compiler/compiler_project_3rd_try
$ ./spellcraft.exe < test.sp
Error: Spell $hp is already conjured!
Error: Undefined spell $unknown.
Warning: Alchemy error! Pours elixir into a mana vessel.
Parsing completed successfully.
```

لینک پروژه:

<https://github.com/o-lieh/SpellCraft-Compiler.git>